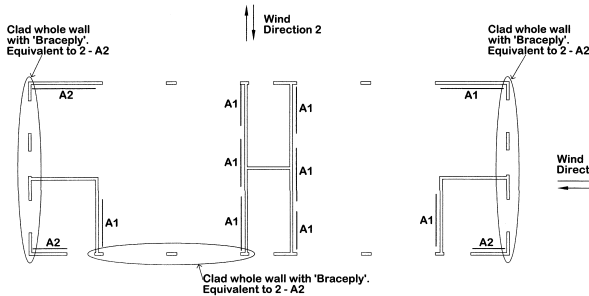


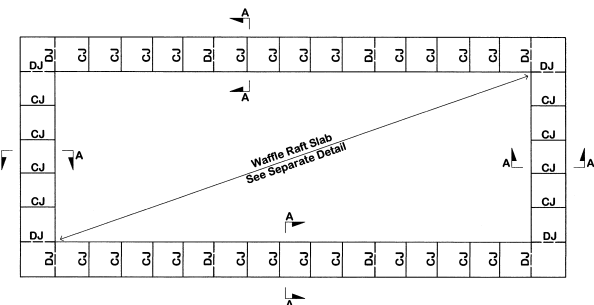
AS4055 - WIND LOADS FOR HOUSING									
WIND BRACING DETAILS - WALLS					TO				
TOPOGRAPHIC CLASSIFICATION					TC2				
TERRAIN CATEGORY					N2				
REGION					1A1, 6A2				
SHIELDING					N2				
WIND CLASSIFICATION					1A1, 6A2				
WIND DIRECTION	FLOOR LEVEL	TYPE	AREA (m ²)	PRESSURE (kPa)	RACKING FORCE (kN)	NUMBER OF BRACINGS REQUIRED	TO BE INSTALLED		
1	GROUND	A	15	0.92	13.8	5A	1A1, 6A2		
2	GROUND	A	36	0.92			1A1, 6A2		

Urban City Certifying
Reference: NS30036/01
Date: 01/05/2023

Accredited Certifier
2573

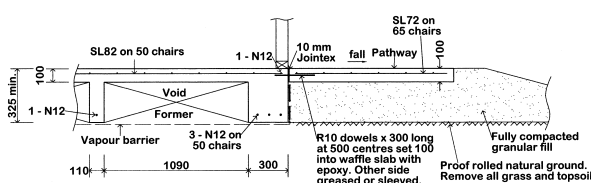


WALL BRACING LAYOUT

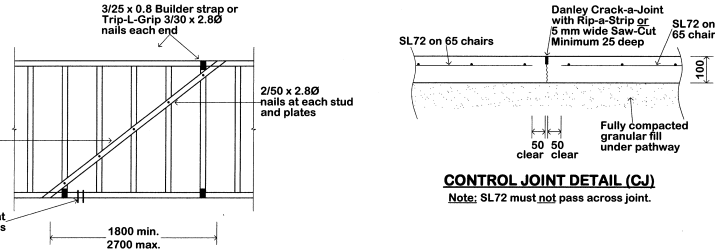


CONCRETE PATHWAY AROUND BUILDING

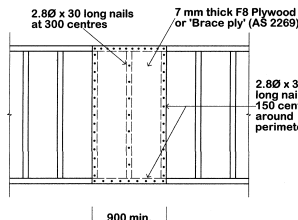
Legend
DJ = Dowelled Expansion Joint. See Detail.
CJ = Control Joint. See Detail.



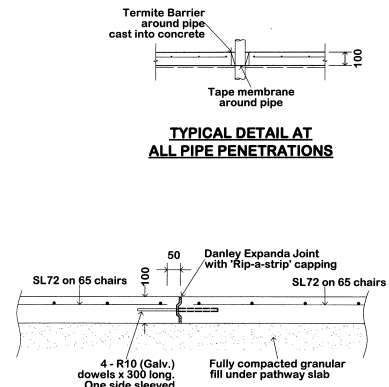
SECTION A-A



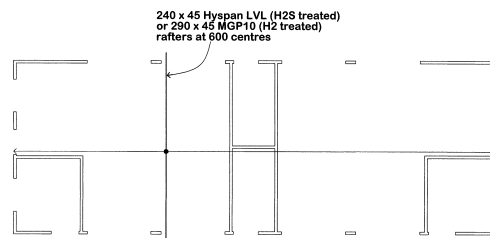
TYPE A1 (3KN)



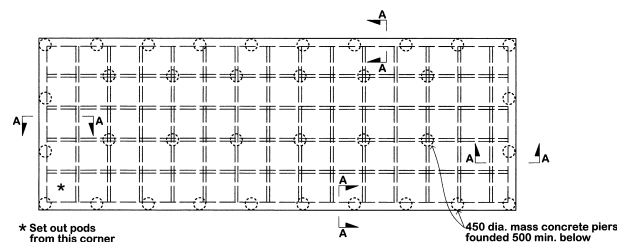
TYPE A2 (3KN)



DOWELLED EXPANSION JOINT DETAIL (DJ)



ROOF RAFTER LAYOUT



WAFFLE SLAB PLAN

100 mm THICK SLAB, SL82 ON 50 mm CHAIRS THROUGHOUT

RESIDENTIAL FOOTINGS AND SLABS

- All residential footings and slabs shall comply with the requirements of A.S. 2870 or as varied on the drawings.
- Footings and slabs are designed for *clad frame* on a class "1" site, as defined in A.S. 2870.
- Site must be adequately drained to prevent all surface runoff and sub-surface seepage from flowing into or under the residence.
- Unless otherwise detailed, the external finished ground level adjacent to the residence shall be 150mm below the adjoining floor level and be graded away from the slab edge to the site drainage system.
- All fill below stiffened raft slabs and slabs on ground shall be fully compacted in accordance with A.S. 2870.
- All fill placed below slab shall be approved material, rolled and compacted in layers by the excavator. Rolled fill shall not exceed 600mm total thickness compacted in layers not more than 300mm thick for sand or 300mm compacted in layers not more than 150mm for other material. Where the fill exceeds the above depths, the affected internal areas of the slab shall be pined under the direction of the Engineer. If
- In the construction of raft slabs and slabs on ground, all grass and topsoil shall be removed in the area of the slab. All tree roots shall be fully removed and backfilled with compacted granular material. Concrete piers where directed by the Engineer shall be provided where tree roots have been removed.

NOTE: These footings may be within the zone of influence of a sewer main. These plans must be approved by Sydney Water prior to commencement of construction.

TIMBERWORK

- All timberwork shall comply with A.S. 1684 and A.S.1720 and those mentioned in A.S.1684 and A.S.1720.
- All timber shall be stress grade MGP10 unless noted otherwise.
- All bolts shall be 12mm diameter steel unless noted otherwise. Bolt holes to be drilled exact size. Washers under heads and nuts to be at least 2.5 times bolt diameter.
- External timber shall be either hardwood durability class 1 or class 2 to AS 1720.2 or impregnated pine grade F7, pressure treated to AS 1604 and re-dried prior to use. Supplementary treatment shall be applied to all cut surfaces. Supply supporting documentation regarding preservation treatment.
- Timber treated in accordance with AS 1604 shall have following hazard level
 - interior above ground - H2
 - exterior above ground - H3
 - exterior in ground - H4
- All timber joints and notches are to be 100mm minimum away from loose knots, severe sloping grain, gum veins or other minor defects.
- Locate a floor joist directly under all walls running parallel to joist.
- Reduce floor joist spacing to 300mm in filled areas.
- Glue laminated timber shall be manufactured in accordance with AS 1328. Members for external use shall be fabricated using resorcinol or phenolic adhesive.
- Camber in beams or rafters shall be as noted on these structural drawings or beam/rafter shall be installed with natural hog up.
- All proprietary fixings shall be installed in accordance with manufacturers recommendations. All metal fixings shall be compatible with timber grades and preservative treatments.
- No penetrations or notches other than those shown on these structural drawings shall be made in any timber members without prior approval of the Structural Engineer.
- Builder to provide termite protection in accordance with AS 3660.1.

WORK HEALTH & SAFETY REGULATION 2011 CLAUSE 295

SAFETY IN DESIGN REPORT

THE STRUCTURAL DESIGN ON THESE DRAWINGS CONTAINS THE HIGH RISK CONSTRUCTION WORK LISTED BELOW. SAFE WORK METHOD STATEMENTS ARE REQUIRED:-

- RISK OF PERSON FALLING MORE THAN 2 METRES
- DEMOLITION WORK
- STRUCTURAL ALTERATIONS
- POWERED MOBILE PLANT

OTHER STRUCTURAL HAZARDS/RISKS THAT MAY OCCUR ARE:-

- FLOORS ARE NOT DESIGNED FOR HEAVY MACHINERY, PLANT AND MATERIALS USED DURING CONSTRUCTION. SAFE LIVE LOADS FOR SUSPENDED FLOORS ARE SHOWN ON THESE DRAWINGS. CONSULT THE EXPOSED ENGINEER WHERE THESE SAFE LIVE LOADS WILL BE EXCEEDED.
- MOST OF THE STRUCTURAL COMPONENTS ON THESE DRAWINGS EXCEED SAFE MANUAL LIFTING BY ONE PERSON. USE SAFE LIFTING METHODS TO AVOID PHYSICAL INJURY.

FLOOR DESIGN LIVE LOADS = 2.0kPa (200kg/m²)

APPROVED FOR CONSTRUCTION		D.J. HALL	
		B. E., M.Eng.Sc., Dip. H. E. (Deflt), M.I.E. Aust. CONSULTING CIVIL & STRUCTURAL ENGINEER NO. 3 KOWLEY STREET, SEVEN HILLS 2147 NSW TEL. NO. - OFFICE - 02 96387973, MOBILE - 046725469 EMAIL ADDRESS - djhall@djhallengr.com@gmail.com	
PROPOSED AMENITIES BUILDINGS			
AT NO. 316 SPRINGWOOD ROAD, YARRAMUNDI			
FOR YMCA YARRAMUNDI			
STRUCTURAL DETAILS			
DATE: 20-9-21	DRAWN BY: LEONARD HALL	SHEET NO. 1 of 1	DRAWING NO. A2112491

SPECIFICATION

GENERAL

- This specification shall be read in conjunction with the structural drawings, architectural drawings and specifications.
- Construction using these structural drawings shall not commence until a Construction Certificate is issued by the Principal Certifying Authority, and only if the structural drawings, are signed approved for construction.
- Setting out dimensions or member sizes must not be scaled from the drawings.
- Any discrepancy or confusion on these drawings must be notified to the Engineer before proceeding further with the work.
- All materials and workmanship shall be in accordance with the requirements of all relevant and current AS Codes and NCC.
- The method of construction and the maintenance of safety during construction is the responsibility of the Builder. If any structural element presents difficulty in respect of constructability or safety, the matter shall be referred to the Structural Engineer for resolution before proceeding with the work. The determination of a safe work method remains the responsibility of the Builder.
- During construction the structure shall be maintained in a stable condition and no part shall be overloaded. Temporary bracing shall be provided by the Contractor in order to keep the building works and excavations stable at all times.
- No changes in any structural element documented in these structural drawings shall be made without reference to the structural engineer. No substitutions shall be made without reference to the structural engineer.
- Proprietary items specified in the structural drawings shall be installed in accordance with manufacturers written recommendations.
- The Builder shall arrange separate certification of any design and construct component by a chartered (NERS registered) structural engineer.
- All architectural fixings (glazing, partitions, ceilings) should allow for the short term and long term movement of structural elements. The Builder should consult the structural engineer for the extent of allowance to be made.
- The structural engineer accepts no responsibility for any works not inspected or not approved by the structural engineer during construction.

FOUNDATION

- All footings and piers shall be founded in even bearing, undisturbed soil or rock of the same type throughout.
- The footings and piers have been designed for a safe bearing capacity of 200kPa. The foundation material shall be approved by the structural engineer or a geotechnical engineer before placing concrete in the footings/piers.
- Unless otherwise shown on these drawings piers are required in soft natural soils, uneven bearing conditions and in filled areas. Unless otherwise specified, piers shall be 400mm diameter mass concrete and spaced at 200mm max. centres. All footings, isolated pads, raft slab edge and internal beams shall be pined. The founding depth of the piers and the adequacy of the foundation material shall be determined on site by the structural or geotechnical engineer.
- Excavations for drainage lines or other services must not effect slab or footing stability. The engineer must be consulted when the trench excavations are within the influence zone of the footings and slabs.
- Foundation excavations shall be kept free of water. If the bearing surfaces deteriorate before concrete is poured, the Contractor shall clean out all loose soil material down to the specified bearing capacity at his own expense.
- Footings shall be located centrally under walls and columns unless noted otherwise on the structural drawings.
- Footings shall be constructed and backfilled as soon as possible following excavation to avoid either softening of the founding material or drying out by exposure.
- The Builder shall ensure the stability of adjacent buildings and services during the excavation for the footings.

CONCRETE

- All workmanship and materials shall be in accordance with current editions of A.S. 3600 except as varied by the contract documents. Ready mix concrete supply shall comply with AS 1379.
- Cement to be type "GP" unless noted otherwise. Concrete components and quality shall be as follows:-

Element	Strength Grade Mpa	Slump mm	Max size Agg	Density kg/m ³
Piers	20	100	20	2400
Slabs	25	100	20	2400

- All concrete shall be vibrated to ensure complete placement into the formwork.
- Concrete shall be cured for 7 days by keeping all exposed surfaces wet with water or another method approved by the engineer.
- No concrete shall be placed until the engineer has inspected and approved the placed reinforcement.
- All concrete used in suspended slabs and beams shall be sampled and tested in accordance with A.S. 1379.
- All reinforcement shall be as follows:-
SYMBOL TYPES
R Structural grade round bars Grade 250 to A.S. 4671
N Deformed bars, normal ductility, Grade 500 to AS4671
SL, RL Deformed wire, square or rectangular mesh, Class L, Grade 500 to AS4671
Note: The number following R & N is bar diameter in millimetres.
8. Clear concrete cover to reinforcement shall be 65mm for footings, 25mm for slab, 50mm for beams and columns or as otherwise noted on the drawings.
9. Splices in the reinforcement shall be made only in the positions shown or as otherwise approved by the engineer.
10. Where transverse bars are not shown provide N12-400. Splice where necessary and lap with main bars for 400mm.
11. Splices in reinforcement shall be Trench Mesh and Fabric-2 Transverse wires plus 25mm, N12-500mm, N16-650mm or as otherwise noted on the drawings.
12. No holes or chases or embedments of pipes other than those shown on the structural drawings shall be made in concrete members without the prior approval of the Engineer.
13. Reinforcement is represented diagrammatically, it is not necessarily shown in true projection.
14. Welding of reinforcement is not permitted unless shown on the drawings or approved by the Engineer.
15. Top and bottom reinforcement in slabs shall be supported in both directions at maximum centres of 600mm for bars 10mm or less, 900mm for bars 12mm and 10mm and 900mm for fabric.
16. Slabs and beams shall bear only on the beams, wall, etc. shown on the drawings. All other building elements shall be kept 15mm clear from soffits of the structure.
17. Vapour barrier shall be high impact resistance polythene sheet with a minimum thickness of 0.2mm. All joints shall be lapped and sealed.
18. No admixtures shall be used in concrete unless approved in writing by the Engineer.
19. Concrete sizes shown do not include thickness of applied finishes. Sizes shall not be changed without the approval of the structural engineer.
20. The surface finish of the concrete shall be as specified on the Architectural drawings.